B.TECH. AI & DATA SCIENCE | SEMESTER V | 2026-27

Smart Lost & Found System

Campus Property Recovery & Management System



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Department of AI & Data Science | RTU Kota
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Project Team & Mentors

PROJECT GUIDE & MENTOR

Er. Ram Babu Buri

Assistant Professor

HEAD OF DEPARTMENT

Dr. Ashok Kajla

Professor, AI & Data Science

MN

Mitesh Nehra

24EARAD102

Student Panel

Lost/found forms, notifications, claim workflow

MS

Mohit Kumar Sharma

24EARAD103

Database & DAO Layer

JDBC manager, schema bootstrap, database.sql

NP

Naitik Pathak

24EARAD106

Admin Panel

Verification queue, desk entry, claimed-items audit

PS

Paramveer Singh

24EARAD119

Testing & Packaging

35-case test suite, build pipeline, portable .exe

PC

Prabhash Kumar Chaudhary

24EARAD121

Authentication & Security

Login, registration, validation, role-based sessions

Problem Statement & Objectives

The Problem

- Informal pinboards and security counters with limited hours
- Scattered social-media groups: high noise, low reach
- No remote search, and a high risk of fraudulent claims
- No audit trail showing items reached the real owner

Our Objectives

1. One central platform to report, track and claim items
2. Role-based access: student self-service vs admin verification
3. Automated rule-based matching of lost and found reports
4. Resilient data: 3NF schema, parameterized SQL, auto bootstrap
5. Portable, zero-install Windows executable

Key Features

Student Self-Service Portal

- Report lost or found items with contact details
- Track status: Pending, Approved, Rejected, Claimed
- In-app approval alerts and claim history

Admin Governance Console

- Verification queue with one-click approve or reject
- Direct desk entry for items handed to staff
- Student directory and claimed-items audit

Smart Matching Engine

- Category check plus title keyword overlap
- Flags possible lost-and-found pairs
- Transparent rules in MatchingService.java

Cloud Connection Lifecycle

- Releases the DB connection on window close
- JVM shutdown hook as a safety net
- Liveness check and timeouts for network drops

Technology Stack

LANGUAGE

Java 17

Core logic, event handling, multithreading

USER INTERFACE

Java Swing & AWT

Custom dark theme via UITheme.java

DATABASE

MySQL 8.0 / Clever Cloud

InnoDB engine, utf8mb4, ACID persistence

CONNECTIVITY

JDBC Type-4

MySQL Connector/J 8.x driver

ARCHITECTURE

Layered + DAO pattern

Presentation, model, service, persistence decoupled

BUILD & PACKAGING

build.bat + jpackage

Portable .exe with embedded Java runtime

15 Java source files

4,500+ lines of Java code

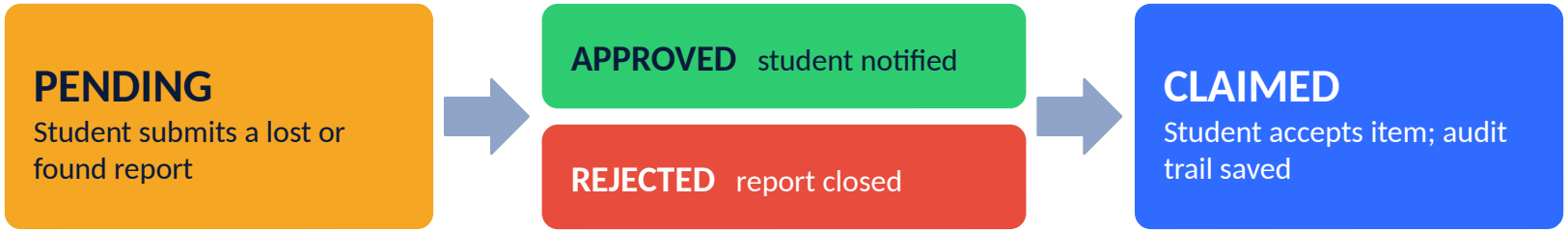
5 architecture layers

Five-Layer System Architecture

1	**Presentation Layer (Swing GUI)**	LoginFrame | RegisterDialog | DashboardFrame | UITheme
2	**Domain Model Layer**	User | LostItem | FoundItem | Claim
3	**Algorithmic Service Layer**	MatchingService (rule-based lexical matching)
4	**Data Access Object (DAO) Layer**	UserDAO | LostItemDAO | FoundItemDAO | ClaimDAO
5	**Persistence & Database Layer**	DatabaseConnection (singleton) > MySQL 8.0 / Cloud DB

Each layer talks only to the layer below it; every SQL call goes through a DAO class.

Item Lifecycle & Role-Based Workflow



Student Portal

- Report lost items
- Report found items
- My Reports with colour-coded status
- In-app notifications and approvals
- Claim history and handover details

Admin Console

- Registered students directory
- Review pending found items
- Direct desk found-item entry
- Review pending lost items
- Claimed items audit and resolution

Database Design (3NF Schema)

users

id (PK)
full_name
email (UK)
role: STUDENT / ADMIN
phone
college_id

lost_items

id (PK)
user_id (FK)
item_name
category
location, lost_date
status

found_items

id (PK)
user_id (FK)
item_name, category
found_location
storage_location
status

claims

id (PK)
lost_item_id (FK)
found_item_id (FK)
claimant_id (FK)
proof_details
claim_status

users 1 : N lost_items, found_items, claims | found_items 1 : N claims | lost_items 1 : N claims (optional link)

Third Normal Form (3NF)

Foreign keys with ON DELETE CASCADE

100% PreparedStatement queries

Auto schema bootstrap + seed data

Smart Matching Engine

1 **Category must match**
Compared case-insensitively, e.g. Electronics = electronics

2 **Title overlap**
One title contains the other, or any word of 3+ letters is shared

= **Result label**
"Possible Match Found" or "No Match Found" shown in the dashboard

WORKED EXAMPLES (ILLUSTRATIVE)

Lost: Black Wallet (Accessories)
Found: Wallet (Accessories)
Title contains "wallet"
MATCH

Lost: Blue Calculator (Electronics)
Found: Calculator Casio (Electronics)
Shared keyword "calculator"
MATCH

Lost: ID Card (Documents)
Found: Black Wallet (Accessories)
Categories differ
NO MATCH

Testing & Project Deliverables

35/35

formal test cases passed

0 defects reported

Coverage: login and roles, validation, reporting, matching, claims, SQL injection, DB cleanup, network recovery

REPRESENTATIVE TEST CASES

TC-03	Admin login opens admin dashboard	PASS
TC-06	Duplicate email registration rejected	PASS
TC-11	Smart match finds candidate pairs	PASS
TC-15	SQL injection input stored as plain data	PASS
TC-17	DB connection closed on window close	PASS
TC-19	Auto-reconnect after network drop	PASS

DELIVERABLES

- Project Report
- SRS (16 FRs)
- IEEE Research Paper
- Demo Video (~10 min)
- 12-Week Dev Log
- Portable .exe

Conclusion & Future Scope

CONCLUSION

- Replaces notice boards and chat groups with one structured, role-based workflow
- Verified approvals, claim tracking and a permanent audit trail
- Secure (parameterized SQL), tested (35/35) and portable (zero-install .exe)

FUTURE SCOPE

- Photo upload and image-based matching
- Email / SMS alerts besides in-app notifications
- Web and mobile front end on the same database

Thank You

Questions & Discussion

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